Exercise 5.3

1. hyperbola 2 circle 3. ellipse 4. circle 5. hyperbola 6. parabola

Exercise 5.4

1. $x-y-3=0, x-9y+13=0$ 2. $10x-3y+32=0,\ 10x-3y-32=0$
3. $(-3,1)$ 4. $x-y+4=0$
5. $x-2y+8=0$ 6. $4x-3y-6=0, 3x+4y-42=0$

Exercise 5.5

1. 8.4 m 2. 26.6 m 3. 3 m
4. $y^2=4.8x,\ 1.3m$ 5. $3.52m, 5.08m$ 6. 90.82m, 148.91m
7. $\frac{2\sqrt{2}}{3}$ 8. $3\sqrt{3}m$ 9. $\tan^{-1}\left(\frac{4}{3}\right)$ 10. $\frac{x^2}{9}-\frac{y^2}{16}=1$

Exercise 5.6

1	2	3	4	5	6	7	8	9	10	11	12	13
(1)	(3)	(4)	(3)	(3)	(1)	(1)	(3)	(2)	(2)	(1)	(4)	(3)
14	15	16	17	18	19	20	21	22	23	24	25	---
(3)	(1)	(4)	(4)	(1)	(1)	(2)	(2)	(3)	(3)	(3)	(2)	---

Exercise 6.1

11. 80 units 12. 69 units
13. $\sqrt{179}, \frac{3}{\sqrt{179}}, \frac{-11}{\sqrt{179}}, \frac{-7}{\sqrt{179}}$ 14. $-96\hat{i}+115\hat{j}+15\hat{k}$

Exercise 6.2

1. 24 2. 720 cubic units 3. -5
4. ± 12 5. $\frac{2\sqrt{3}}{5}$ 6. coplanar 7. 2

Exercise 6.3

1. (i) $-2\hat{i}+14\hat{j}-22\hat{k}$ (ii) $22\hat{i}+14\hat{j}+2\hat{k}$ 5. -74
7. $l=0,\ m=10,\ n=-3$ 8. $\theta=\frac{\pi}{3}$

Exercise 6.4

1. $\left(\vec{r}-\left(4\hat{i}+3\hat{j}-7\hat{k}\right)\right)\times\left(2\hat{i}-6\hat{j}+7\hat{k}\right)=\vec{0}$, $\frac{x-4}{2}=\frac{y-3}{-6}=\frac{z+7}{7}$
2. $\vec{r}=\left(-2\hat{i}+3\hat{j}+4\hat{k}\right)+t\left(-4\hat{i}+5\hat{j}-6\hat{k}\right)$, $\frac{x+2}{-4}=\frac{y-3}{5}=\frac{z-4}{-6}$
3. $\left(\frac{32}{3},0,\frac{47}{3}\right), (0,16,-11)$

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